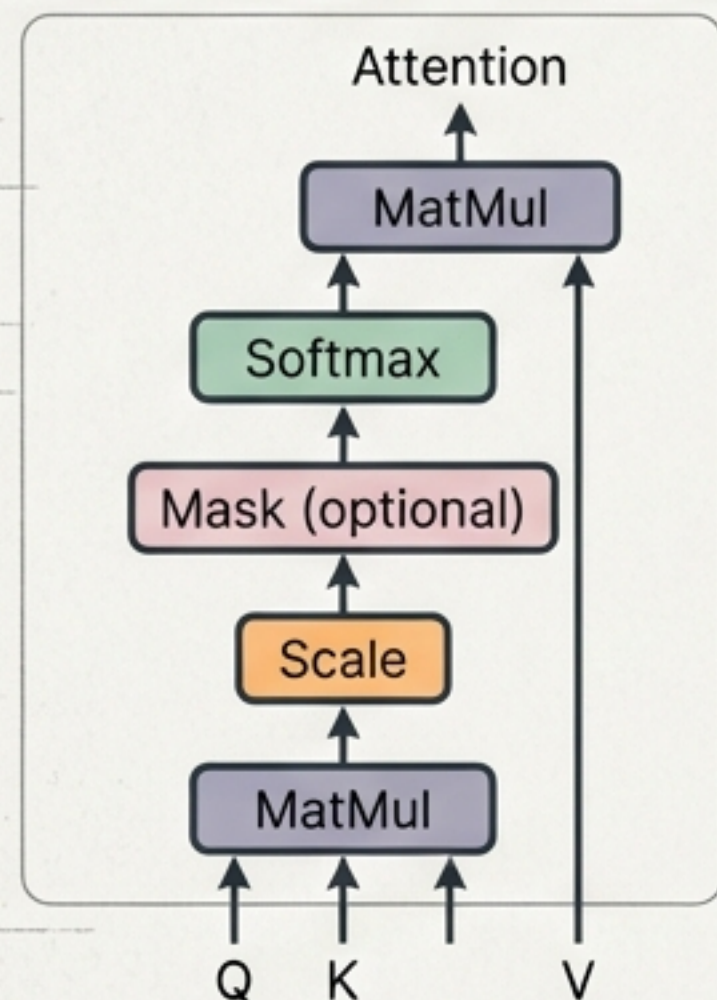


Multi-Head Attention: "Attention Is All You Need"

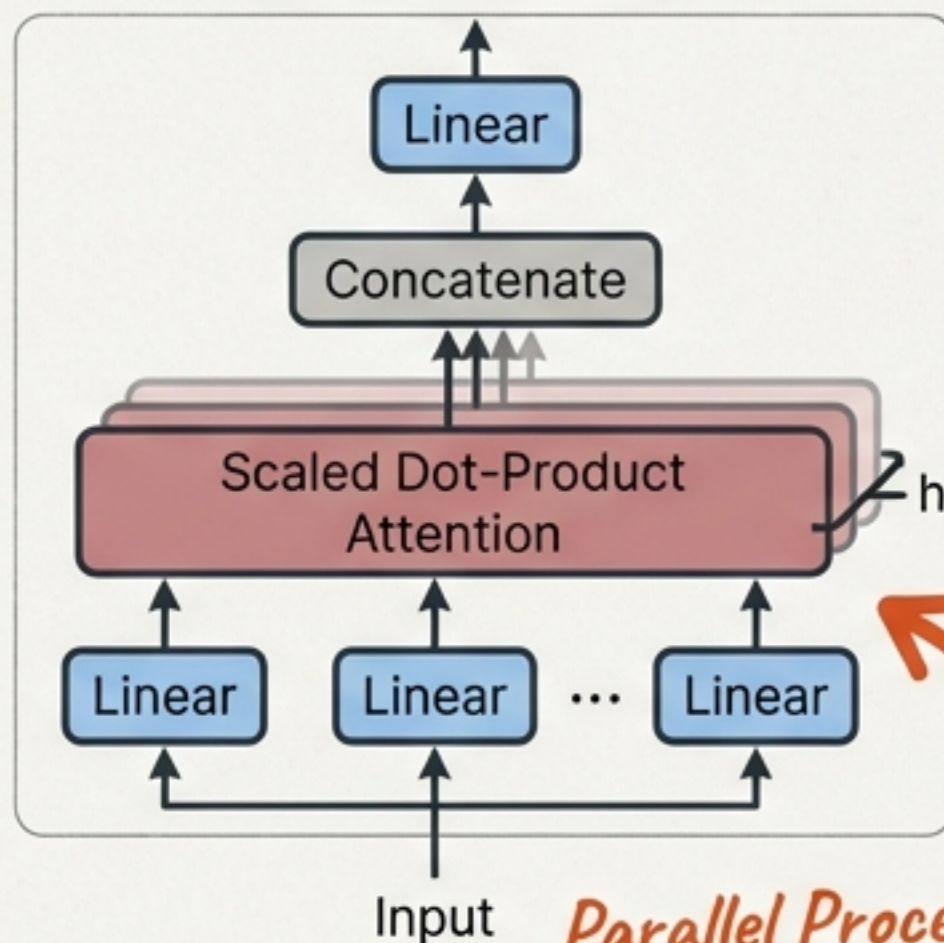
The Technical Foundation

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Scaled Dot-Product Attention



Multi-Head Attention



*Parallel Processing:
No waiting in line.*

ARCHITECTURE: Transformer (Vaswani et al., 2017)

ARCHITECTURE: Transformer (Vaswani et al., 2017)

COMPONENT: Multi-Head Attention Unit

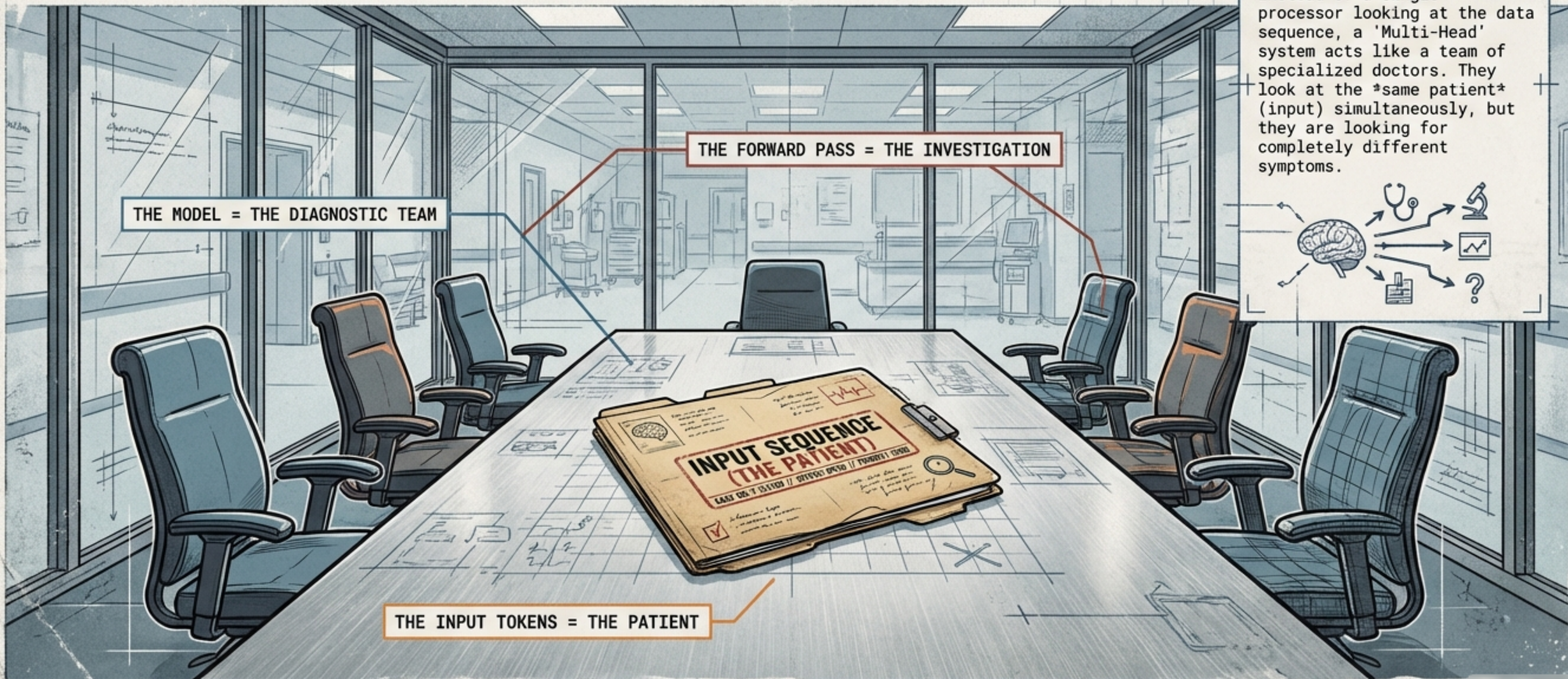
HEAD COUNT: 8 to 16 (Typical)

MATRICES: Query (Q), Key (K), Value (V)

The Mechanism: The model does not focus on a single aspect of the input. Instead, it splits the input embedding into multiple "heads." Each head projects the input into independent Query, Key, and Value matrices. This allows the model to attend to information from different "representation subspaces" (e.g., grammar, semantic meaning, position) simultaneously.

The Analogy: Differential Diagnosis

Reframing Processing as Investigation



Parallel Representation Subspaces

CONFIDENTIAL

Action: Attends to subject-verb agreement and structural dependencies. Ignores emotion.

CONFIDENTIAL

[illegible]

Action: Attends to physical anomalies and object connections. Links "Patient" to "Surgery".

CONFIDENTIAL

Action: Attends to emotional weight. If the patient says "I'm fine" but mentions "divorce," this head flags the contradiction.

Clinical Notes

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This image shows a blank, aged, cream-colored page, likely an endpaper or flyleaf of a book. The paper has a slightly textured appearance with some minor creases and discoloration, characteristic of old paper. The left edge of the page shows the binding of the book.

1

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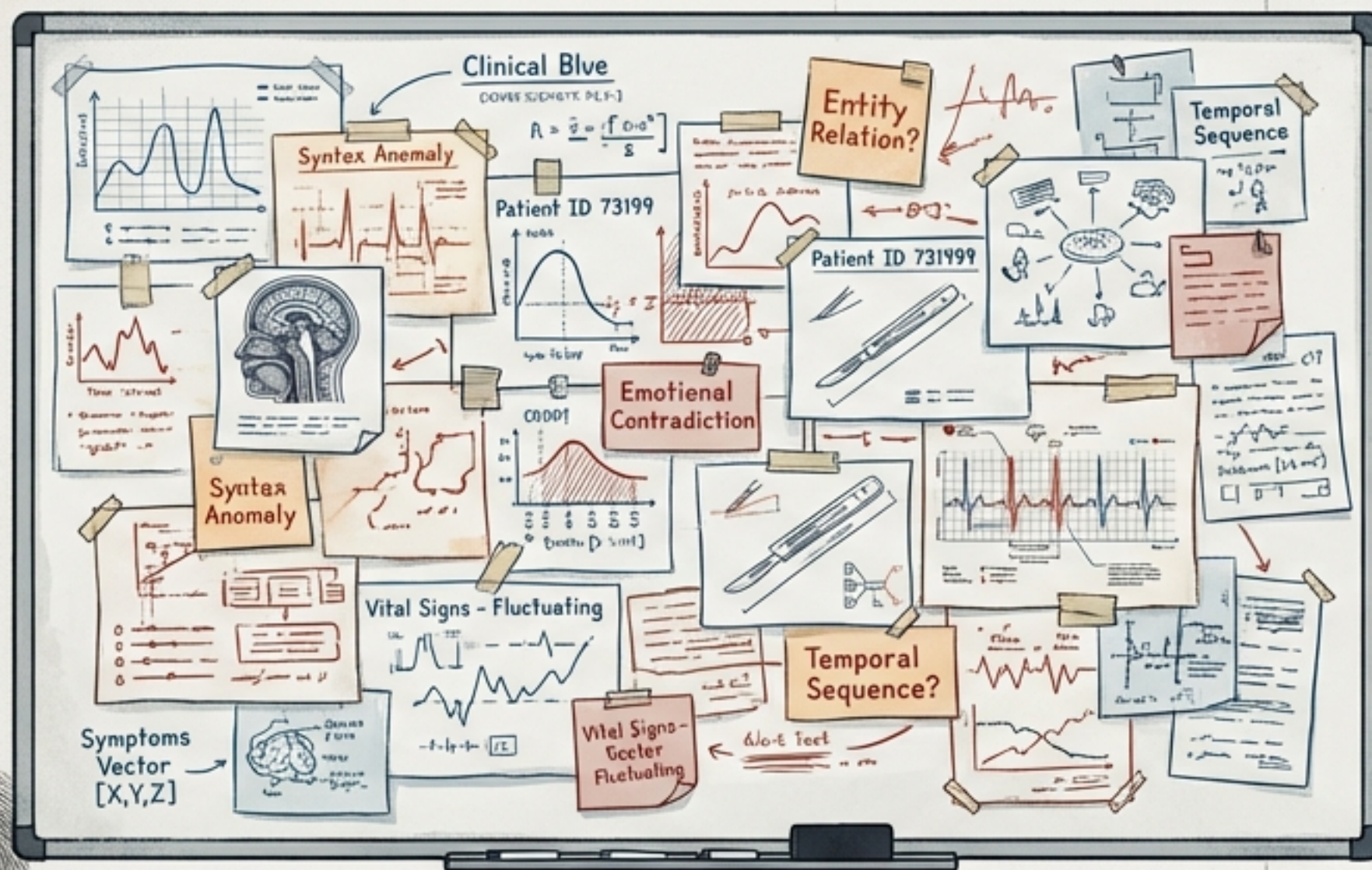
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NotebookLM

Concatenation & The Linear Layer

The Whiteboard and The Boss



CONCATENATION

CONCATENATION:
Aggregation of all findings into a single master sequence.



THE LINEAR LAYER

THE LINEAR LAYER:
The brilliant synthesist extracting the final truth.

The Mechanism:

1. Concatenation (The Whiteboard): The independent vectors (findings) from all heads are combined into one long master sequence. No information is discarded yet; it is simply aggregated.
2. The Linear Layer (House): The final projection layer acts as the brilliant synthesist. It looks at the messy, combined data and extracts the single, final truth.

OUTPUT VECTOR:
"It's a parasite"
(Next Token Prediction)

Avoiding Tunnel Vision

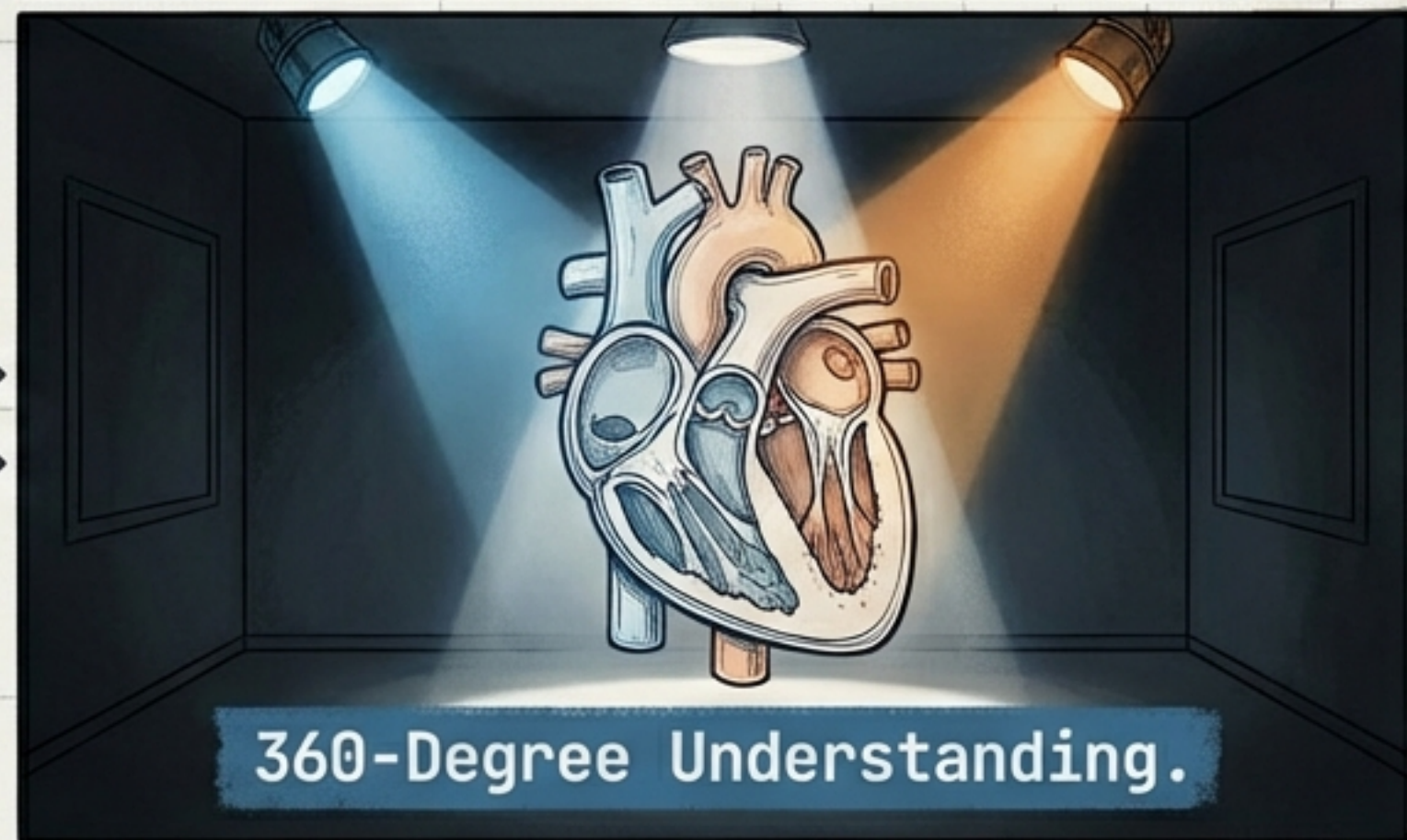
Why Multi-Head Architecture Matters

SINGLE-HEAD RISK



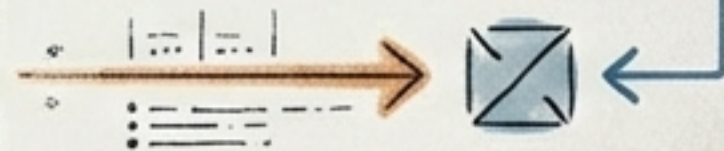
A single head might get distracted by one obvious symptom and miss the underlying cause.

MULTI-HEAD ROBUSTNESS



By forcing the model to "obsess" over different parts of the data (syntax, sentiment, entities) simultaneously, the Transformer gains a robust view of the truth.

"The team generates the perspectives;
the architecture makes the diagnosis."



Clinical Notes

*Notes for patient
presentation - medical
outcomes*

deriving

*Davidson
Bantz H."*